On Consistency between General Relativity and Galaxy Rotation Curve

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Abstract

In this paper, we found a metric which can derive galaxy rotaion curve without dark matter assumption, and related it to CMB temperature.

Introduction

In 1978, Rubin showed some galaxies rotate in constant speed. However, current models either assume dark matter which blows up to infinity, or modify gravity which can't apply to classical ones. Hence we try finding a model without dark matter but still within GR.

Kepler's Curve

First recall how does Newton's gravity predict: From Schwarzschild metric

$$ds^2 = \left(1 - \frac{r_s}{r}\right)(cdt)^2 - \frac{dr^2}{1 - \frac{r_s}{r}} - r^2 d\Omega^2$$

derive geodesic equations with $v^2 = c^2$ and $r' = 0$, we found

$$r\frac{d\theta}{dt} = \sqrt{\frac{GM}{r}}$$

Hence the pattern is replacing a metric to end with

$$r\frac{d\theta}{dt} = c\beta$$

where $c\beta$ is the rotation speed.

Galaxy Rotation Curve

The usual approach is back to Newton's approximation

$$ds^2 = \left(1 + \frac{2V}{c^2}\right)(cdt)^2 - \frac{dr^2}{1 + \frac{2V}{c^2}} - r^2 d\Omega^2$$

imposing consistency and hence dark matter. But we insist that radial part is classical, then there is no choice

$$ds^2 = \left(1 - \frac{r_s}{r}\right)(cdt)^2 - \frac{dr^2}{1 - \frac{r_s}{r}} - e^{2h} d\Omega^2$$

where $h = h(r)$. Imposing consistency, we found

$$e^{2h} = \frac{r r_s}{\beta^2}$$

Cosmic Microwave Background

I know that r behaviour instead r^2 is absurd, but a recent research on CMB showed

$$kT = \frac{\hbar c}{4\pi\sqrt{r_s R_H}}$$

where r_s is chosen to m_p and $R_H = \dfrac{c}{H_0}$. The similar "square root of two distance" pattern makes us confide that our metric wants to tell us something, just we don't know.

Reference

[1] Vera C. Rubin & W. Kent Ford, Jr. & Norbert Thonnard "Extended rotation curves of high-luminosity spiral galaxies. IV. Systematic dynamical properties, Sa → Sc"
[2] Espen Gaarder Haug & Stéphane Wojnow "How to predict the temperature of the CMB directly using the Hubble parameter and the Planck scale using the Stefan-Boltzman law"